

Experiment 2 — FPGA-Based Smart Energy Monitoring System

RTL module: Smart_Energy_Monitoring Testbench: tb_Smart_Energy_Monitoring Simulation duration: 60 ns

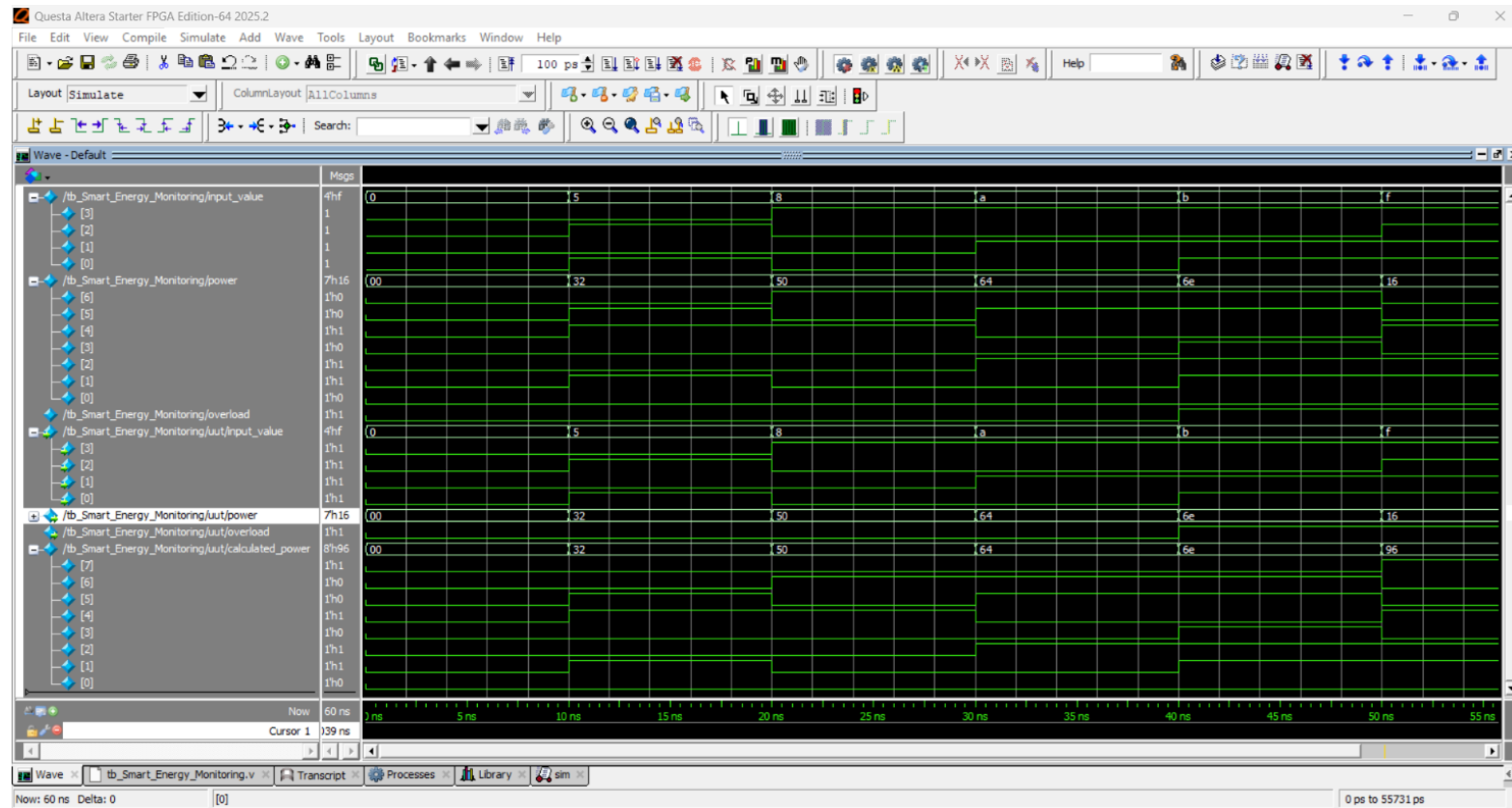
Simulation Objective

The simulation verifies the calculated power output and overload detection. Calculated power is input_value × 10, while overload becomes active above 100.

Verification Sequence

Time	Input	Power	Overload
0–10 ns	0	0	0
10–20 ns	5	50	0
20–30 ns	8	80	0
30–40 ns	10	100	0
40–50 ns	11	110	1
50–60 ns	15	150	1

Waveform Evidence — The screenshot below is the captured Questa waveform for this experiment.



Result: The waveform completes the 60 ns test sequence and demonstrates the expected logical transitions for input_value, power, overload. The simulation stops normally at the end of the testbench.

Conclusion

Simulation verification completed successfully with the DUT and corresponding testbench loaded in Questa. The observed transitions are consistent with the intended RTL logic.